Speckle Interferometry of WDS 21555+1053 (BU 75 AB), WDS 19594+3206 (A 378) and WDS 20213+02550 Binary Star Systems

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Abstract

Astrometric measurements of the physical systems WDS 21555+1053, WDS 19594+3206, and WDS 20213+0250 were performed using speckle interferometry. The measurements used the 17″ Boyce Astro Research Observatory's Plane Wave BARON telescope. The measurements yielded a mean separation distance and mean separation angle of 1.139″ ± 0.005″ and 28.376° ± 0.07°, respectively, for WDS 21555+1053, and a mean separation distance and angle of 0.939″ ± 0.006″ and 292.201° ± 0.587 for WDS 19594+3206. The measurement for WDS 21555+1053 agrees well with the published sixth orbital catalog. The measurement for WDS 19594+3206, combined with other recent observations, indicates that the system's orbit may be more eccentric than the historical ephemeris allows; additional observations are needed. For WDS 20213+0250, the mean separation distance and separation angle are observed to be 1.25″ ± 0.008″ and 40.028° ± 0.455°. The measurement for WDS 20213+0250 also supports the published orbit.

1. Introduction

In this work, we measure the separation angle and distance of the binary star systems WDS 21555+1053, WDS 19594+3206, and WDS 20213+0250. We then contrast the historical orbit published by the Sixth Orbital Catalog with an optimized orbit based on our measurements. The specifications of the three systems are given in Table 1 below, including the constellation in which the binary stars are located, their distance from Earth (given in light years), their orbital period (given in years), the magnitude of their primary and secondary stars, and their spectral type (Stelle Doppie 2018).

Table 1. Specifications of the binary stars studied in this work

	WDS 21555+1053 BU 75 AB	**WDS 19594+3206 A 378**	**WDS 20213+0250 HLD 158**
Constellation	Pegasus	Cygnus	Aquila
Distance (lyr)	155.86	155.17	1437.01
Period (yr.)	158.5	591.3	232.4
Mag (primary)	8.40	8.48	9.71
Mag (secondary)	8.56	8.98	9.78

Spectral type	G5 (yellow)	G5 (yellow)	G0 (yellow)

The system WDS 21555+1053 is well-constrained and categorized as grade 2 in the Sixth Catalog of Orbits of Visual Binary Stars. The system WDS 19594+3206 is classified as grade 5. There have been very few measurements on this system, with the last measurement being made in 2021. This measurement was performed using speckle interferometry (Beavers et al. 1989) and showed a slight departure from the published orbit. Its orbital elements are, therefore, not well constrained, suggesting a need for continued observation. The system WDS 20213+0250 is categorized as grade 3. The most recent measurement was made in 2021.

The paper is organized into five sections. Section 2 describes the equipment and methods used in observing our star systems. Section 3 contains the data collected from our observations. Section 4 analyzes the data, and section 5 presents our conclusions.

2. Equipment and Methods

Measurements for this paper were collected using the Boyce Research Initiative and Education Foundation's BARON telescope (BRIEF 2024) located at the Sierra Remote Observatory in California, United States (Figure 1). The 17-inch PlaneWave CDK telescope has an aperture of 431 mm and a focal length of 2930 mm. Imaging was taken on June 21st, 23rd, and 25th, 2024 (average epoch 2024.479) through Sloan r filters with an exposure time of 40 milliseconds. Calibration images were captured by directing the telescope to a star cluster and generating images with 30-second exposure times. Reference stars were chosen from the calibration images using the Plate Solve 3.80 algorithm (Dave Rowe), which generated plate rotation angle and image scale parameters.

Figure 1: BRIEF's 17-inch BARON telescope at the Sierra Remote Observatory.

Speckle interferometry was used via software developed by Dave Rowe called Speckle Toolbox (Harshaw 2017) as a method of resolving the images by minimizing atmospheric turbulence. Large samples (500 images) of short-exposure (40 msec) images were taken and overlaid to reduce atmospheric effects and obtain a higher resolution. Turbulent airflow in Earth's atmosphere distorts imaging of celestial objects, causing them to appear blurred. Speckle Toolbox is used to perform a Fourier mathematical analysis on the short-exposure images to isolate signals from the target stars from the atmospheric disturbance. Speckle Toolbox applied its bi-spectrum phase analysis algorithm to the short-exposure images and the data obtained from the calibration image to calculate the separation angle and distance. Figure 2 shows an example of an unresolved image (left) and a resolved image (right) of the primary and secondary star of the binary system WDS 19594+3206 after speckle interferometry. The more refined image allows for more accurate measurement of the stars' positions.

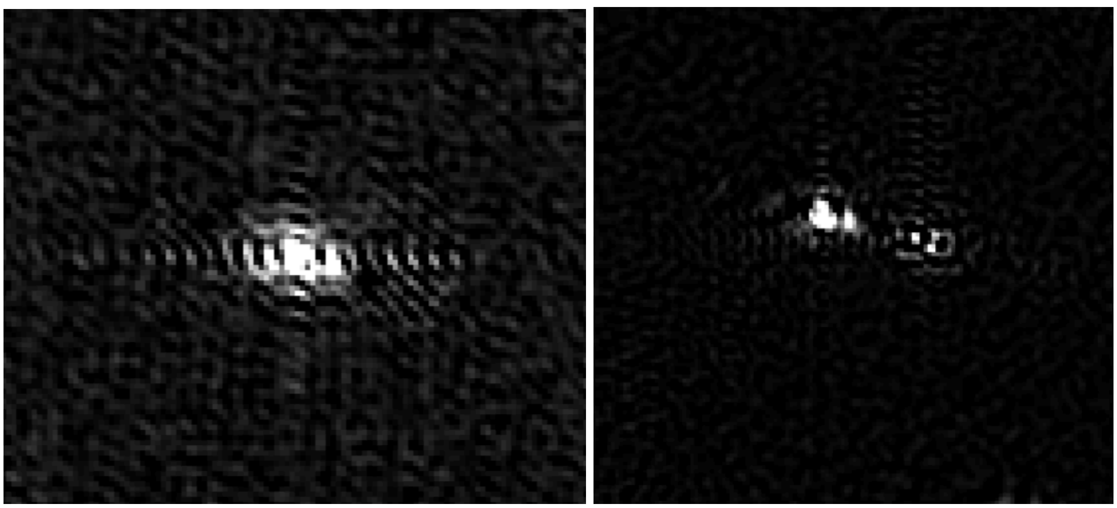

Figure 2: WDS 19594+3206 speckle unresolved (left) and the resolved (right) image

Figure 3 shows an example of the bi-spectrum photometry analysis for the binary system WDS 19594+3206. The primary star is circled in green, and the secondary star is circled in magenta.

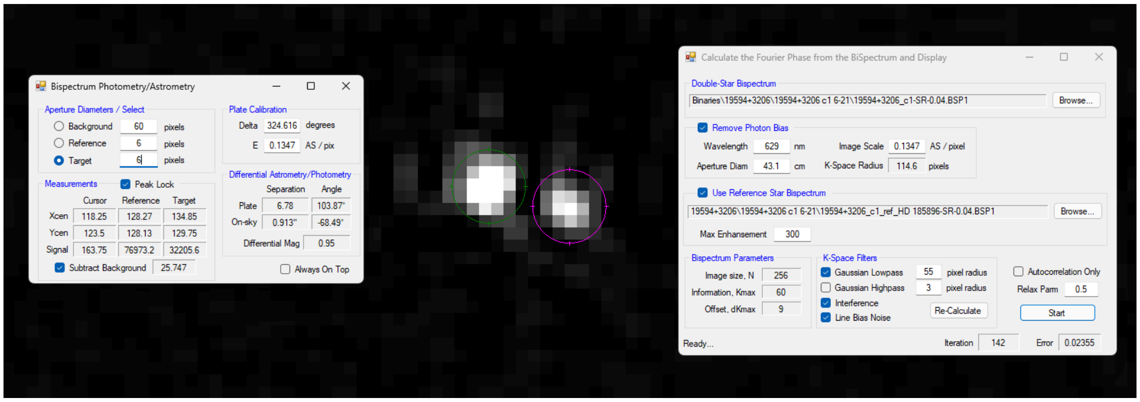

Figure 3: Screenshot of the data reduction on WDS 19594+3206

3. Data

The data was collected over three nights (June 21, 2024; June 23, 2024; and June 25, 2024). A Sloan r filter was used to collect five hundred images with an exposure time of 0.04 s for each cycle. The data collected from two nights for WDS 21555+1053 and WDS 20213+02550, and from three nights for WDS 19594+3206 are shown in Table 2. WDS 21555+1053 used reference star HD 210717, WDS 19594+3206 used reference star HD 194260, and WDS 20213+02550 used reference star HD 193374.

Table 2. Data collected for WDS 21555+1053, WDS 19594+3206, and WDS 20213+02550

WDS	21555+1053		**Discoverer**	BU 75			
	Reference Star:	HD 210717					
Date	**Epoch**	**Observatory**	**Filter**	**Exposure (msec)**	**(S) Star (R) Ref.**	**Images per Cycle**	**Number of Cycles**
6/21/24	2024.474	BARON	Sloan r	40	S	500	2
6/21/24	2024.474	BARON	Sloan r	40	R	500	2
6/25/24	2024.485	BARON	Sloan r	40	S	500	1
6/25/24	2024.485	BARON	Sloan r	40	R	500	1

WDS	19594+3206		**Discoverer**	A 378			
	Reference Star:	HD 195594					
Date	**Epoch**	**Observatory**	**Filter**	**Exposure (msec)**	**(S) Star (R)**	**Images per Cycle**	**Number of Cycles**

					Ref.		
6/21/24	2024.474	BARON	Sloan r	40	S	500	1
6/21/24	2024.474	BARON	Sloan r	40	R	500	1
6/23/24	2024.479	BARON	Sloan r	40	S	500	1
6/23/24	2024.479	BARON	Sloan r	40	R	500	1
6/25/24	2024.485	BARON	Sloan r	40	S	500	1
6/25/24	2024.485	BARON	Sloan r	40	R	500	1

WDS	20213+0250		**Discoverer**	HLD 158			
	Reference Star:	HD 193374					
Date	**Epoch**	**Observatory**	**Filter**	**Exposure**	**(S) Star**	**Images per**	**Number**
				(msec)	**(R) Ref.**	**Cycle**	**of Cycles**
6/21/24	2024.474	BARON	Sloan r	40	S	500	2
6/21/24	2024.474	BARON	Sloan r	40	R	500	2
6/23/24	2024.479	BARON	Sloan r	40	S	500	1
6/23/24	2024.479	BARON	Sloan r	40	R	500	1

The reduction results shown in Tables 3, 4, and 5 are averages from multiple observations reduced by two different people to produce more accurate results. The speckle reduction technique described above in section 2 was used to analyze the data.

Table 3. Measured mean separation angle (θ) and separation distance (ρ) for WDS 21555+1053

WDS 21555+1053 BU 75 AB				
Epoch	Measurement	Theta (degrees)	Rho (arcseconds)	Delta Magnitude (Sloan R)
2024.479	Mean	28.376	1.139	0.61
	Standard Deviation	0.172	0.013	0.054
	Standard Deviation of the mean	0.07	0.005	0.022
Ephemeris		28.58	1.124	

Table 4. Measured mean separation angle (θ) and separation distance (ρ) for WDS 19594+3206

WDS 19594+3206 A 378				
Epoch	Measurement	Theta (degrees)	Rho (arcseconds)	Delta Magnitude (Sloan R)
2024.479	Mean	292.201	0.939	0.949
	Standard Deviation	1.439	0.014	0.539
	Standard Deviation of the mean	0.587	0.006	0.022
Ephemeris		291.47	0.92	

Table 5. Measured mean separation angle (θ) and separation distance (ρ) for WDS 20213+0250

WDS 20213+0250 HLD 158				
Epoch	Measurement	Theta (degrees)	Rho (arcseconds)	Delta Magnitude (Sloan R)
2024.476	Mean	40.028	1.25	0.419
	Standard Deviation	1.115	0.019	0.069
	Standard Deviation of the mean	0.455	0.008	0.028
Ephemeris				

The mean separation distance and separation angle for WDS 21555+1053 are observed to be 1.139″ ± 0.005″ and 28.376º ± 0.07º, respectively. For WDS 19594+3206, the mean separation distance and separation angle are observed to be 0.939″ ± 0.006″ and 292.201º ± 0.587º. For WDS 20213+0250, the mean separation distance and separation angle are observed to be 1.25″ ± 0.008″ and 40.028º ± 0.455º.

Figure 4 depicts the historical orbital plot of WDS 21555+1053 from the Sixth Catalog of Visual Binary Star set against our observed result (red dot, BARON2024). This system is categorized as a grade 2 orbit, and our results are in agreement. The blue points represent observations analyzed via speckle interferometry, and the green points represent historical observations analyzed with various other methods. Figures 4, 5 and 6 were created using the Desmos Graphical Analysis tool (Desmos 2023) and the historical data from the Sixth Orbital Catalog.

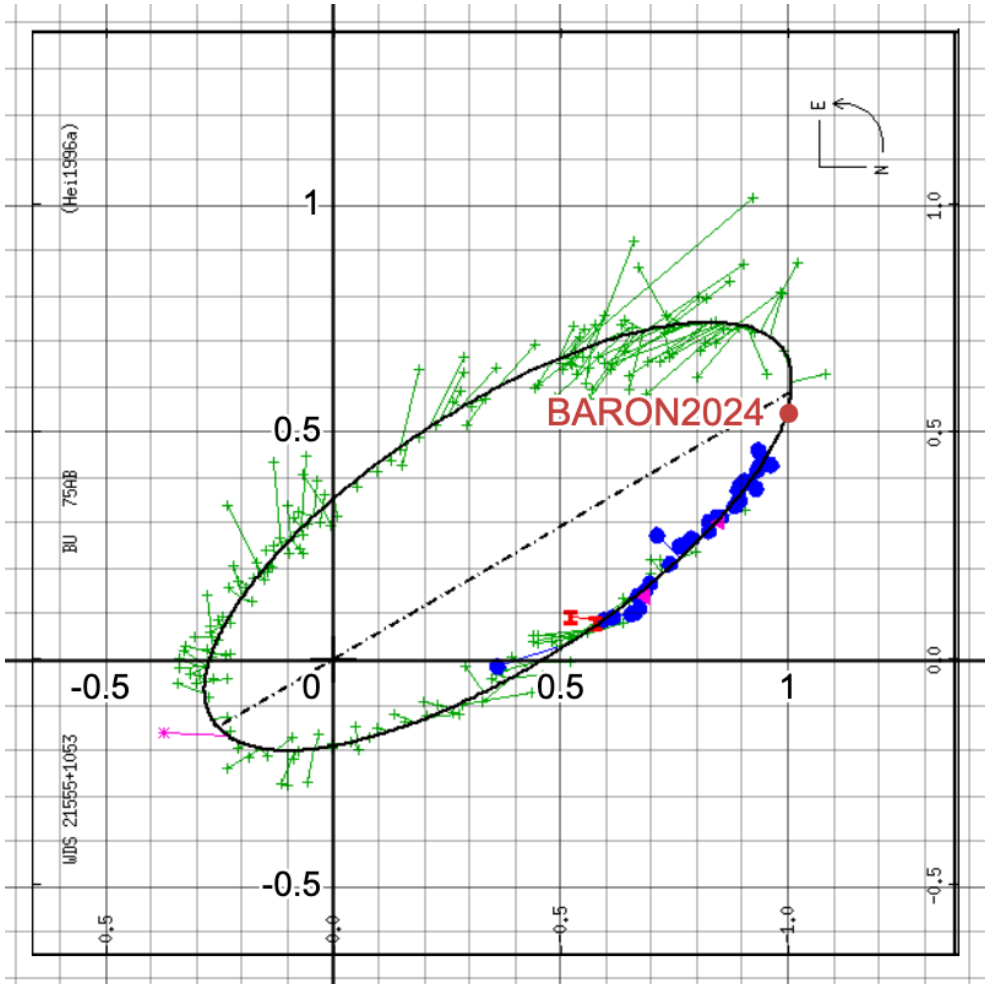

Figure 4: Apparent orbital plot of WDS 21555+1053 with historical data from the sixth orbit catalog. The red solid point (BARON2024) is the observation from the present work.

Figure 5 depicts the historical orbital plot of WDS 19594+3206 from the Sixth Catalog of Visual Binary Star set against our observed result (red dot, BARON2024). This system is categorized as a grade five orbit. The blue points represent observations analyzed via speckle interferometry, and the green points represent historical observations analyzed with various other methods. Also shown is data (WSI2024) measured in 2021 by Matson *et al* (Matson 2024) using speckle interferometry. Most of the speckle data points, including our present observation (red dot, BARON2024) show a slight departure from the sixth catalog orbit.

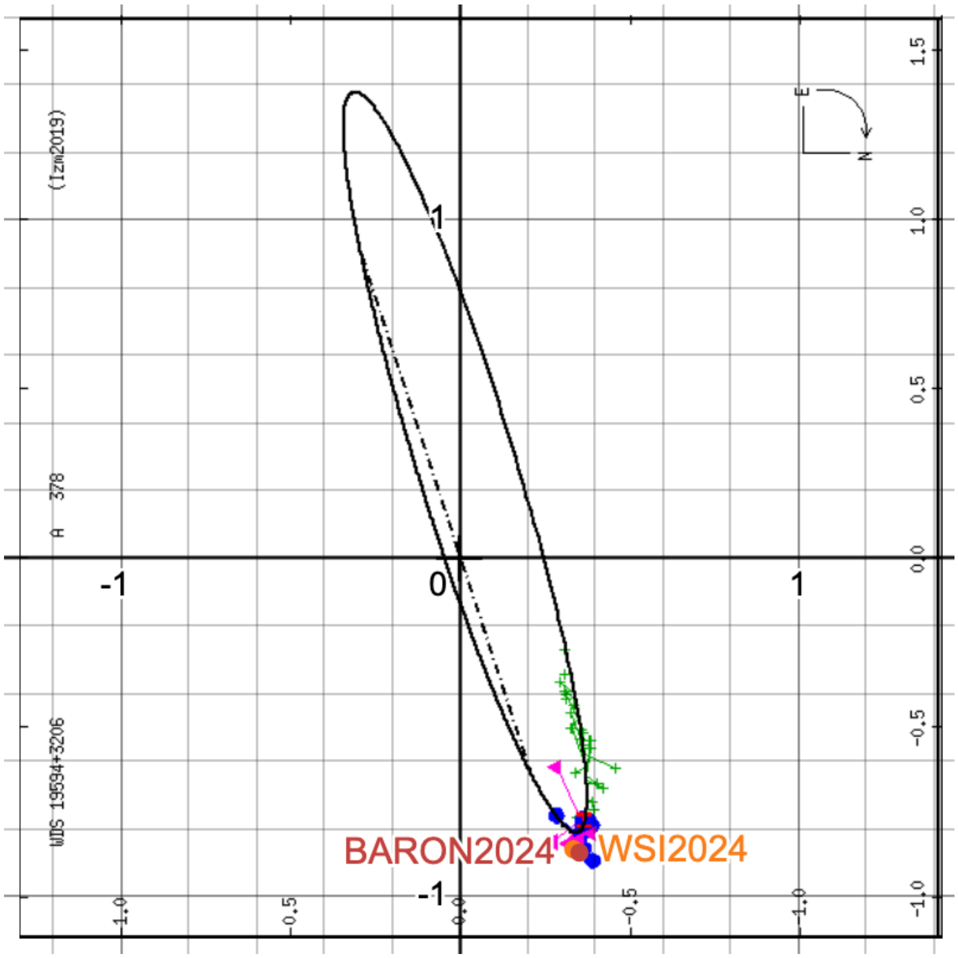

Figure 5: Apparent orbital plot of WDS 19594+3206 with historical data from the sixth orbit catalog. The red solid point (BARON2024) is the observation from the present work.

Figure 6 depicts the historical orbital plot of WDS 20213+0250 from the Sixth Catalog of Visual Binary Star set against our observed result (red dot, BARON2024). This system is also categorized as a grade five orbit. The blue points represent observations analyzed via speckle interferometry, and the green points represent historical observations analyzed with various other methods. Also shown is the data (WSI2024) measured in 2021 by Matson *et al* (Matson 2024) using speckle interferometry. All speckle data points, including our present observation (red dot, BARON2024) indicate that **the orbit is well constrained.**

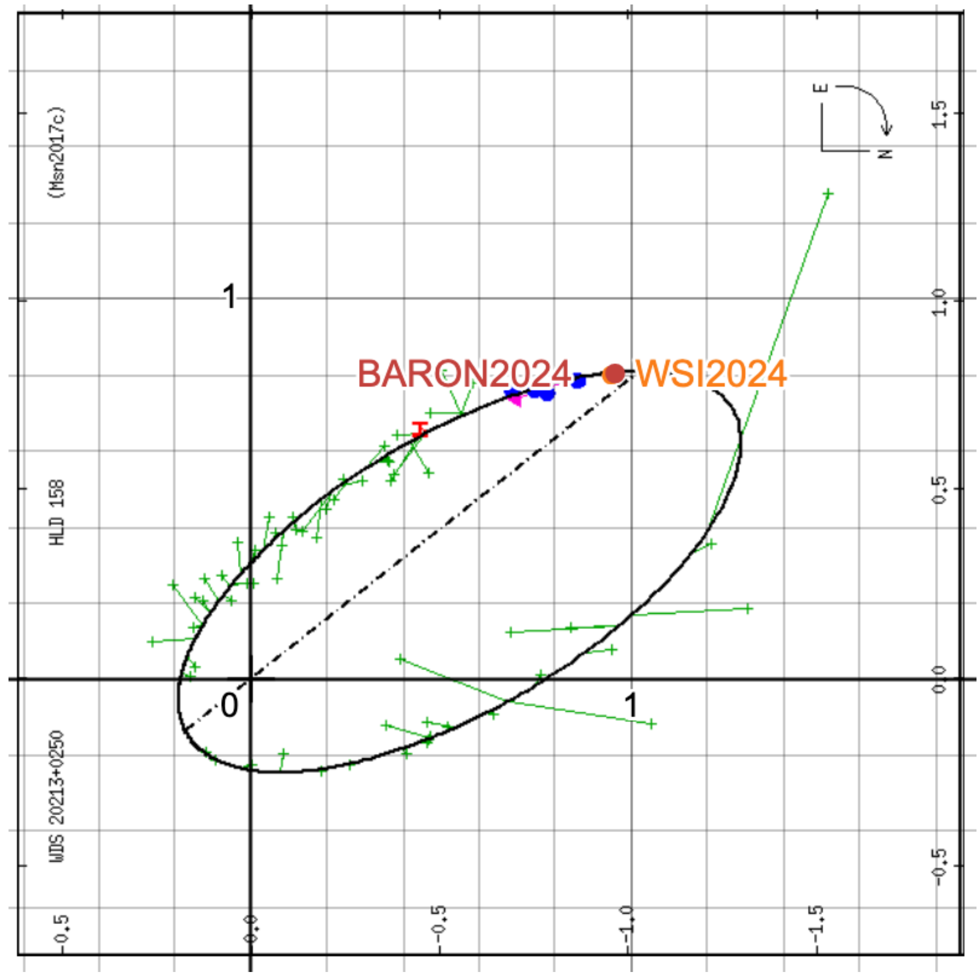

Figure 6: Apparent orbital plot of WDS 20213+0250 *with historical data from the sixth orbit catalog. The red solid point (BARON2024) is the observation from the present work.*

4. Discussion

We optimized the orbit of WDS 19594+3206 to better fit the more recent speckle interferometry points. The Desmos graphic analysis of Genet *et al* (2024) uses the following equations for an elliptical orbit not centered on the origin and with its major axis not coincident with the x-axis: x = (h + a cos t) (cos q) + (k + b sin t) (- sin q); and y = (h + a cos t) (sin q) + (k + b sin t) (cos q), where a is the semi-major axis; b is the semi-minor axis; q is the angle from the x-axis to the ellipse major axis; h is the offset of the ellipse center from the primary star; k is the y offset of the center of the ellipse from the primary star; and t is a parameter that traces out the ellipse from 0 to 360º.

Using the orbit optimization software of Speckle Toolbox by Rowe et al, we generated a new apparent ellipse using the above equations. The dashed orbit in Figure 7 represents the new ellipse.

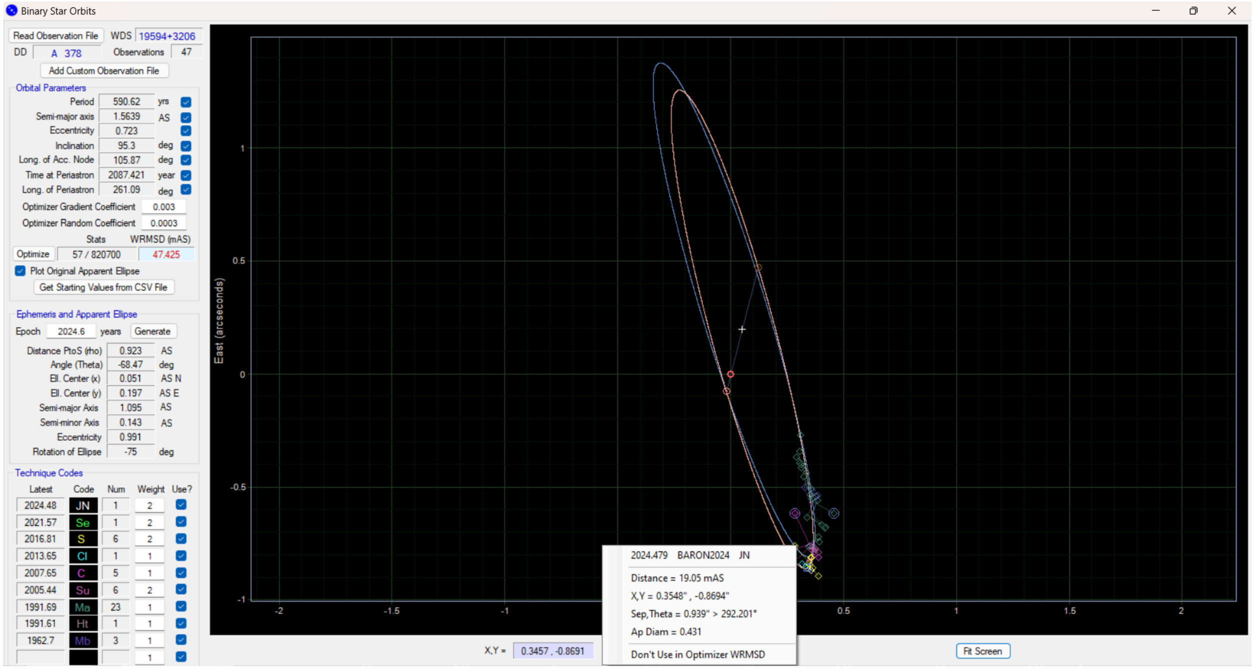

Figure 7: Apparent orbital plot of WDS 19594+3206 with historical data from the sixth orbit catalog. The red solid point (BARON2024) is the observation from the present work. The dashed curve is representative of a new orbit.

Table 5 shows the values for the five ellipse parameters for the original (solid) WDS apparent ellipse and the new (dashed) apparent ellipse shown in Figure 5.

Table 5. Ellipse parameter values for WDS 19594+3206 new orbit and sixth orbit catalog original ellipse.

Parameter	**Name and Units**	**Sixth Orbit**	**New**
a	Semi-major axis (″)	1.141	1.095
b	Semi-minor axis (″)	0.144	0.143
h	Primary star x offset	0.015	0.051
k	Primary star y offset	0.284	0.197
q	The angle of the major axis (deg.)	-69.07	-68.47

Table 6 shows the orbital elements corresponding to the published orbit in the sixth orbital catalog (solid orbit in Figure 7) and as reported by Izmailov *et a*l. in 2019. The table also shows the orbital elements obtained by optimizing the orbit using the more recent speckle interferometry data. The WRMSD in the table refers to the Weighted Root Mean Standard Deviation obtained from the orbit optimization. Continued observation of this binary system may help refine the orbital elements of this system further.

Table 6. Orbital elements for the published orbit in the sixth orbital catalog and the new orbit for WDS 19594+3206

Orbital Elements	Published Orbital Elements (Izm2019) (solid orbit in Figure 7)	New Orbital Elements (Dashed orbit in Figure 7)
	WRMSD (mAS) = 72.203	WRMSD (mAS) = 47.425
Period (yrs.)	591.3	590.62
Semi-major axis (arcsec)	1.5739	1.5639
Eccentricity	0.7096	0.723
Inclination (deg)	95.41	95.3
Long. Of Acc. Node (deg)	108.1	105.87
Time at Periastron (year)	2060.824	2087.421
Long. Of Periastron (deg)	256.25	261.09

5. Conclusions

WDS 21555+1053 and WDS 20213+0250 agree with the WDS predicted orbit published in the Sixth Catalog of Visual Binary Stars. However, the results for WDS 19594+3206 show a slight departure from the WDS-predicted orbit, indicating an orbit that is more eccentric than previously thought. Continued measurement of this system should help constrain the orbit and its orbital elements.

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